

Ergocalciferol from mushrooms or supplements consumed with a standard meal increases 25-hydroxyergocalciferol but decreases 25-hydroxycholecalciferol...

Few foods contain ergocalciferol or cholecalciferol. Treatment of mushrooms with UV light increases ergocalciferol content and could provide a dietary source of vitamin D. We evaluated the impact of consuming UV-treated white button mushrooms (*Agaricus bisporus*) on the vitamin D status of healthy adults. Thirty-eight volunteers were randomized to 4 treatments consumed with a standard meal for 6 wk: the control (C) group received untreated mushrooms providing 0.85 µg/d ergocalciferol (n = 10); groups M1 and M2 received UV-treated mushrooms providing 8.8 (n = 10) and 17.1 µg/d (n = 9), respectively; and the supplement (S) group received purified ergocalciferol plus untreated mushrooms, providing a total of 28.2 µg/d (n = 9). Serum total 25-hydroxyvitamin D [25(OH)D] and 25-hydroxyergocalciferol [25(OH)D₂] were 83 ± 38 and 2.4 ± 2.0 nmol/L, respectively, at baseline (mean ± SD). At wk 6, 25(OH)D₂ had increased and was higher in all treatment groups than in the C group, whereas 25-hydroxycholecalciferol [25(OH)D₃] had decreased and was lower in the M2 and S groups than in the C group. Increases in 25(OH)D₂ for groups C, M1, M2, and S were 1.2 ± 5.2, 13.8 ± 7.3, 12.7 ± 3.7, and 32.8 ± 3.3 nmol/L and decreases in 25(OH)D₃ were -3.9 ± 16.3, -10.4 ± 6.4, -20.6 ± 14.6, and -29.5 ± 15.9 nmol/L, respectively. Concentrations did not change in group C. In summary, ergocalciferol was absorbed and metabolized to 25(OH)D₂ but did not affect vitamin D status, because 25(OH)D₃ decreased proportionally.